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(54) **HYBRID BRIDGE TRANSFER SWITCH SYSTEM AND METHODS**

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H03K 17/56 (2006.01)

(52) **U.S. Cl.**
CPC ***H02J 9/068*** (2020.01); ***H03K 17/56*** (2013.01)

(58) **Field of Classification Search**
CPC H02J 9/068; H03K 17/56
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2009/0072623 A1* 3/2009 Liao H02J 9/061 307/65
2014/0039711 A1* 2/2014 Divan G06Q 30/0279 700/291
2021/0273440 A1* 9/2021 Kang H01L 21/67
2022/0247167 A1* 8/2022 Song H02H 1/0007

FOREIGN PATENT DOCUMENTS

WO 2023114669 A1 6/2023

* cited by examiner

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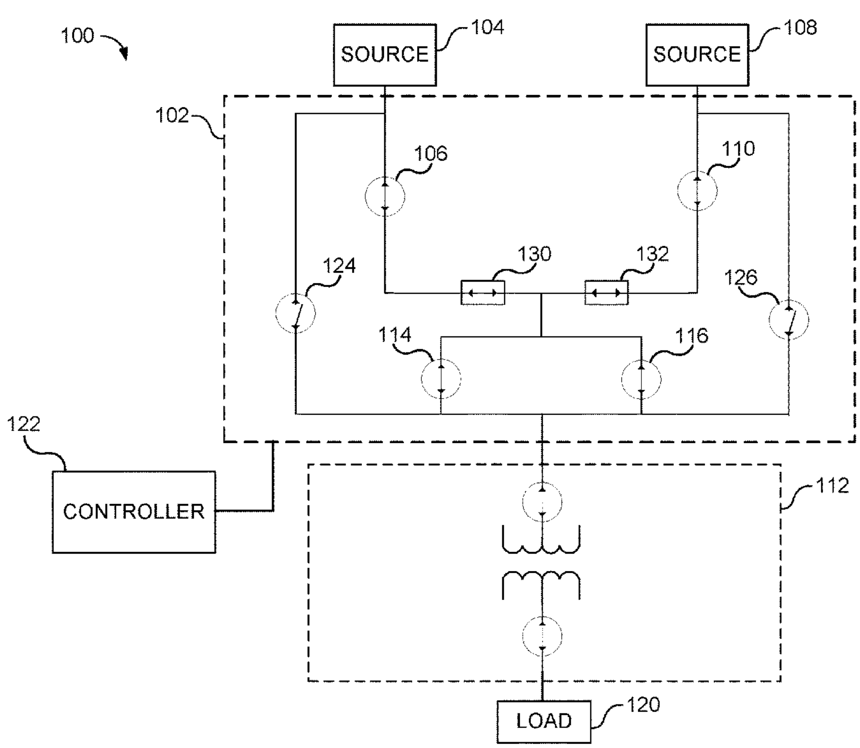
(57) **ABSTRACT**

A transfer switch for hybrid switching between a first power source and a second power source, the transfer switch includes a first switch and a second switch, the first and second switch connecting a load to a first power source and a second power source, respectively. The first switch includes at least one first solid state switching device (SSSD) pair. The second switch includes at least one second SSSD pair. The at least one first SSSD pair includes a different type of switching devices relative to the type of switching devices of the at least one second SSSD pair. Moreover, a controller may be configured to selectively cycle the at least one first SSSD pair and the at least one second SSSD pair on and off to electrically couple one of the first power source and the second power source to the load.

20 Claims, 6 Drawing Sheets



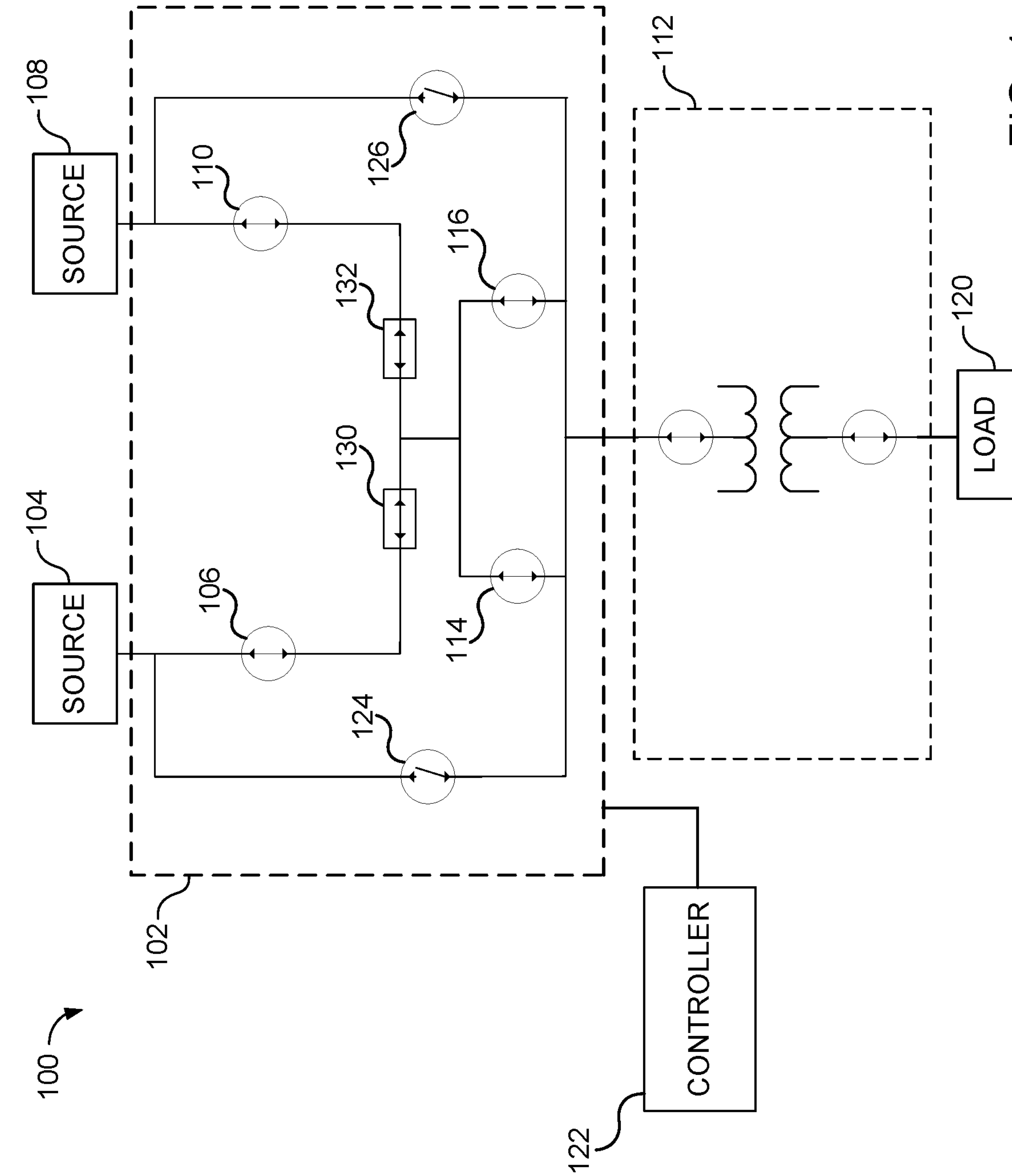

100
102
SOURCE
104
106
SOURCE
108
110
112
114
116
LOAD
120
CONTROLLER
122
124
126
130
132

FIG. 1

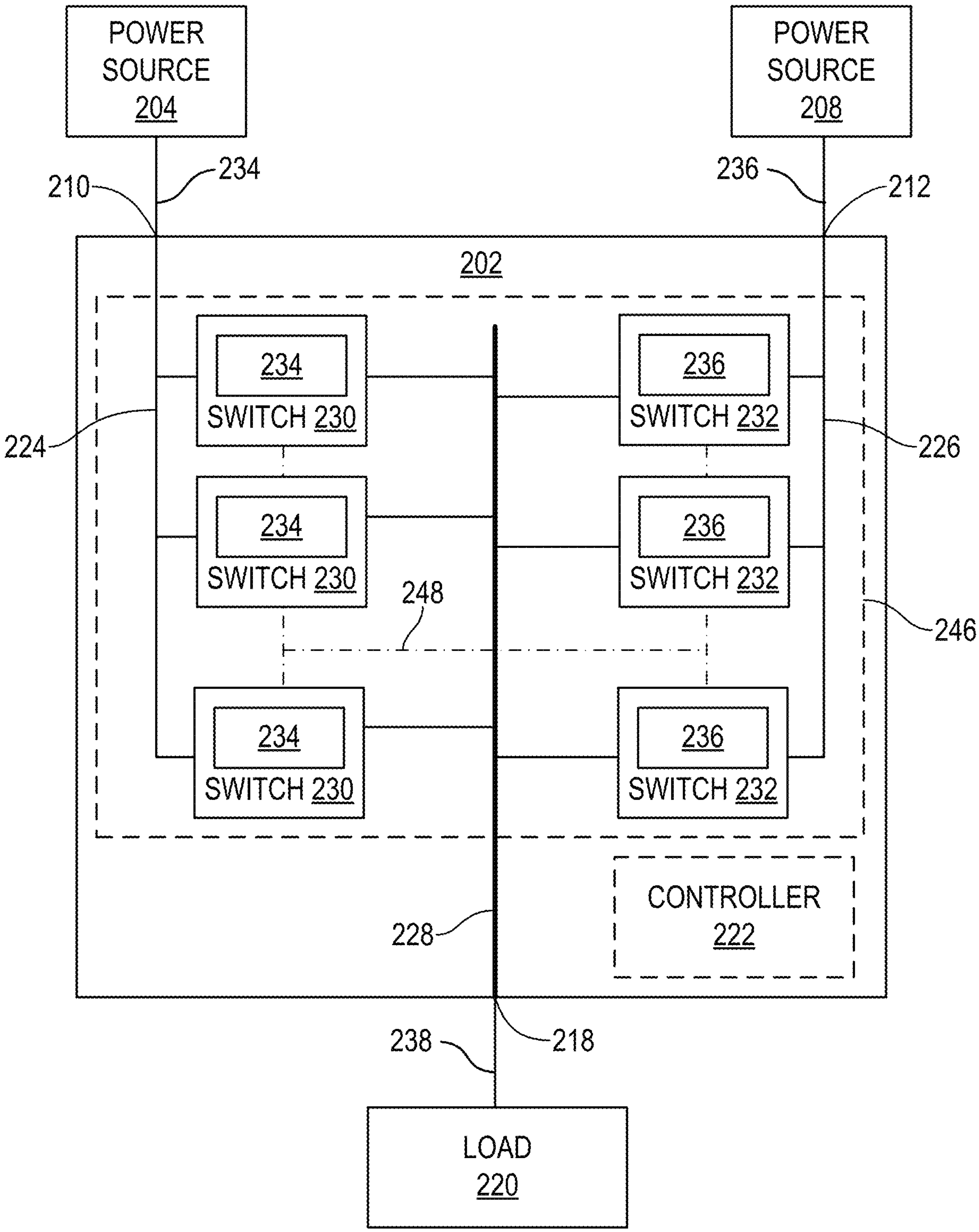

POWER SOURCE 204
POWER SOURCE 208
210
234
236
212
202
234
SWITCH 230
234
SWITCH 230
234
SWITCH 230
236
SWITCH 232
236
SWITCH 232
236
SWITCH 232
224
226
248
246
CONTROLLER 222
228
238
218
LOAD 220

FIG. 2

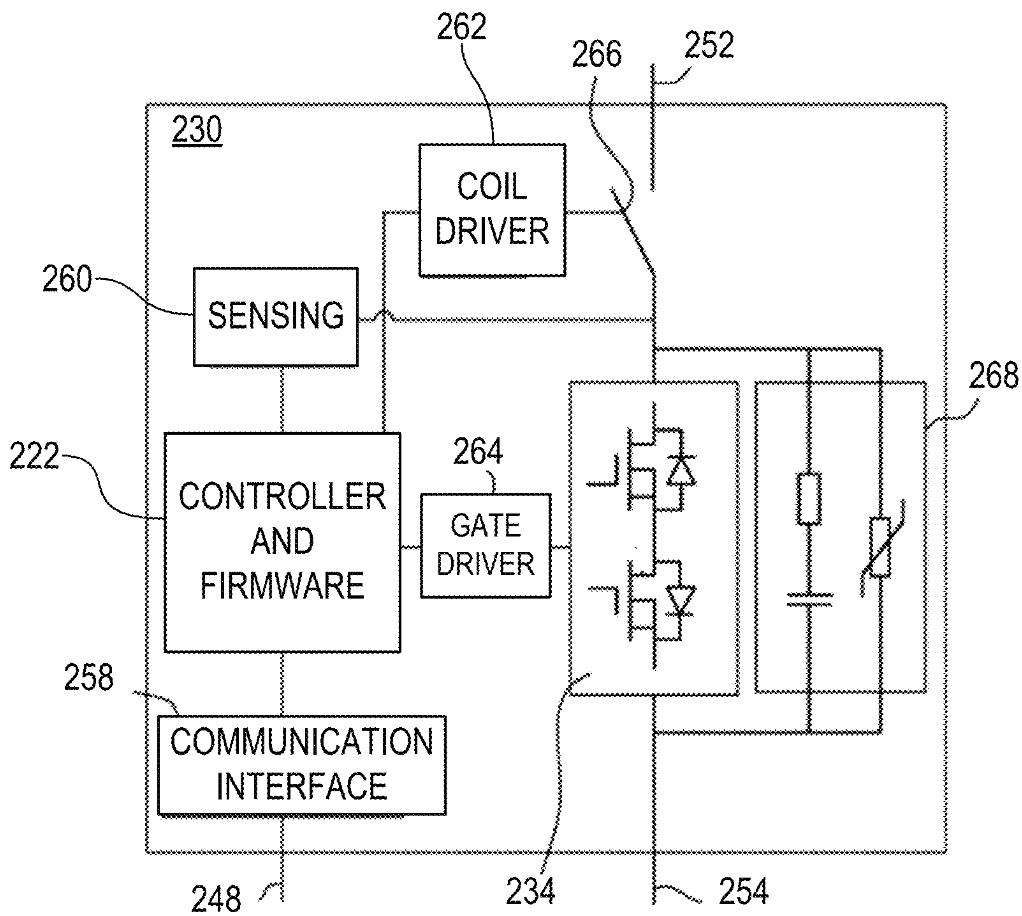

262
266
252
230
COIL DRIVER
260
SENSING
268
222
CONTROLLER AND FIRMWARE
264
GATE DRIVER
258
COMMUNICATION INTERFACE
248
234
254

FIG. 3

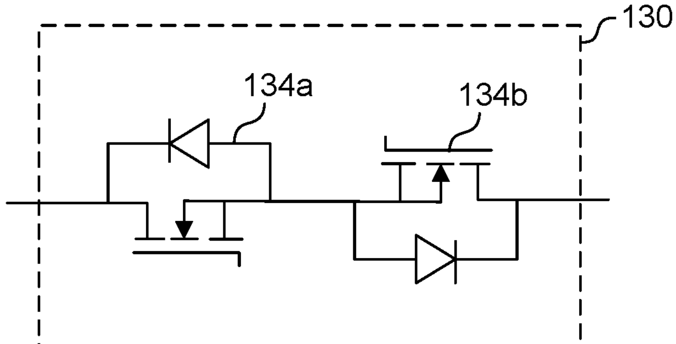


FIG. 4

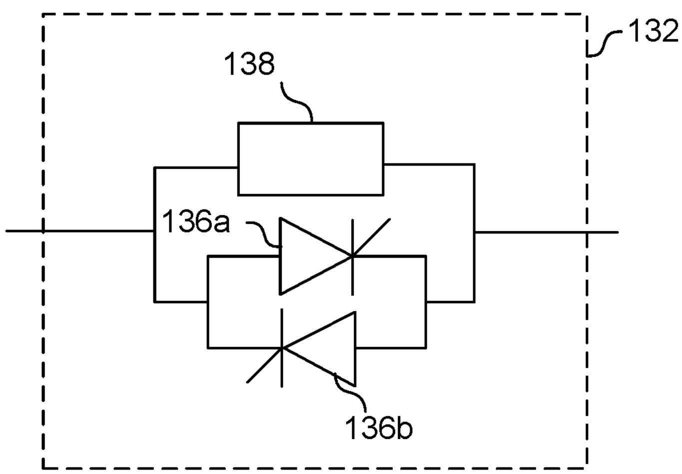


FIG. 5

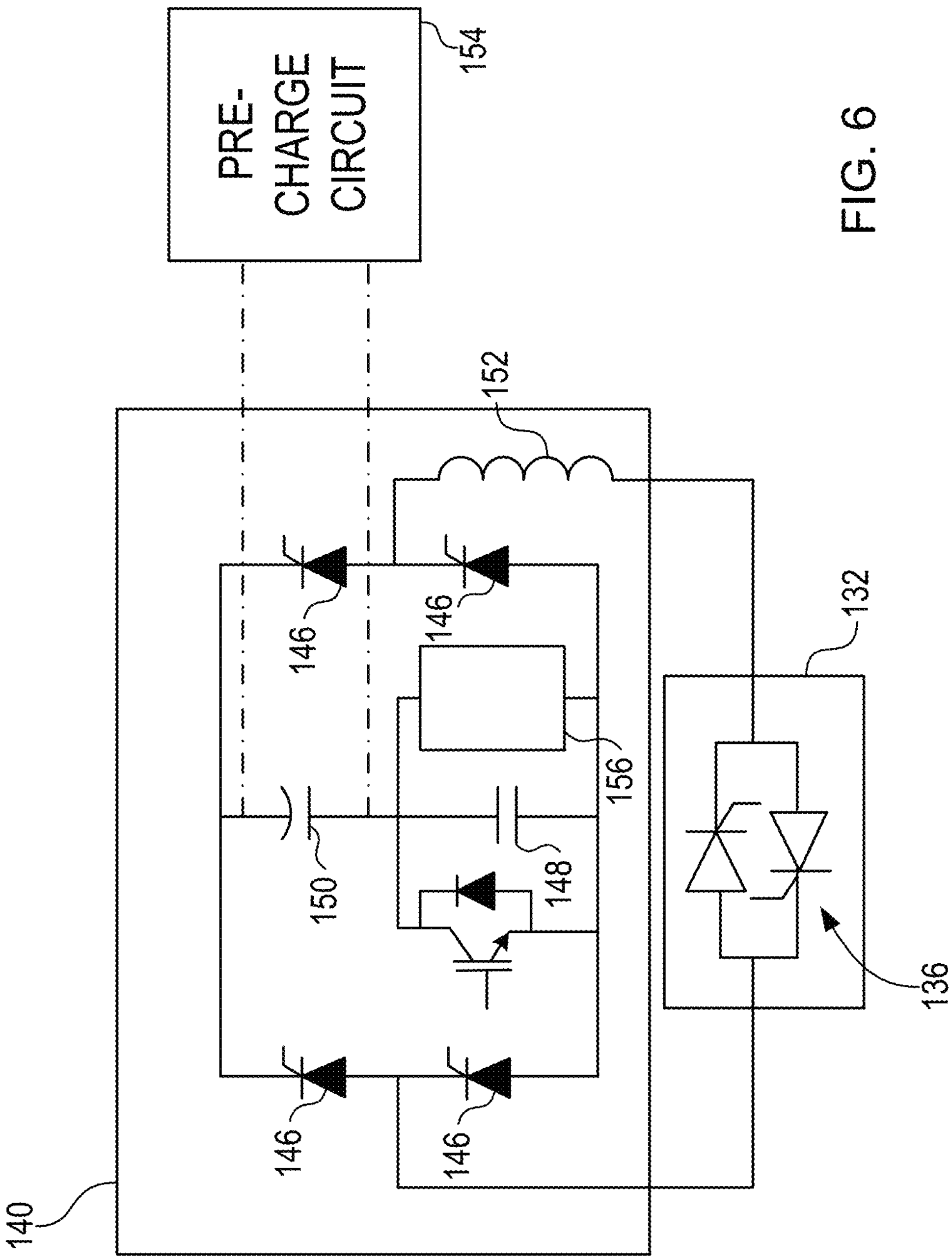

PRE-
CHARGE
CIRCUIT
154
152
146
146
146
146
156
150
148
132
136
140

FIG. 6

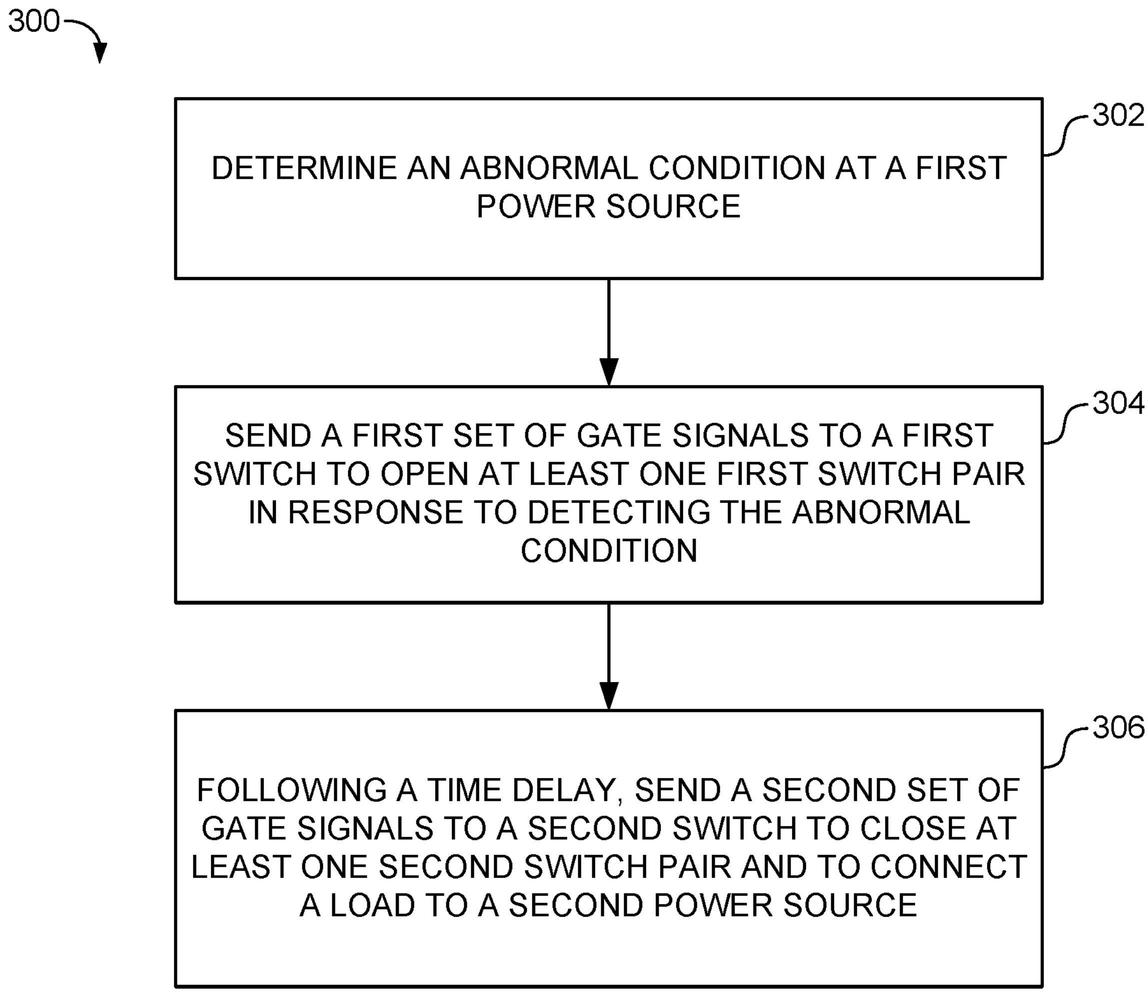

300
DETERMINE AN ABNORMAL CONDITION AT A FIRST POWER SOURCE
302
SEND A FIRST SET OF GATE SIGNALS TO A FIRST SWITCH TO OPEN AT LEAST ONE FIRST SWITCH PAIR IN RESPONSE TO DETECTING THE ABNORMAL CONDITION
304
FOLLOWING A TIME DELAY, SEND A SECOND SET OF GATE SIGNALS TO A SECOND SWITCH TO CLOSE AT LEAST ONE SECOND SWITCH PAIR AND TO CONNECT A LOAD TO A SECOND POWER SOURCE
306

FIG. 7

HYBRID BRIDGE TRANSFER SWITCH SYSTEM AND METHODS

FIELD

The present disclosure relates to the field of transfer switches. And, more particularly, to improving reliability against electrical current instability in transfer switches.

BACKGROUND

Transfer switches are typically used to control the electrical power supply to critical electrical components such as, for example, data centers where a constant, high-quality electrical supply is required. Transfer switches couple electrical load(s) to two electrical power sources. Typically, the electrical power sources may be a primary grid-tied power source and an alternate source such as, for example, electrical generators or a renewable power source. When the electrical power supplied from the primary electrical power source is interrupted, or when fluctuations are detected at the primary electrical power source, the transfer switch performs a switching operation and transfers connection of the electrical load(s) to the secondary electrical power source, where the electrical loads(s) may remain connected until such time as the primary power source stabilizes and the electrical load(s) may be reconnected to the primary electrical power source.

SUMMARY

An objective for ensuring the health of transformers is to maintain a continuous power supply to downstream electrical loads such as, for example, data center applications. When a transformer is initially energized or reenergized using a transfer switch after a brief interruption, it can draw a high inrush current from the system due to residual flux in the core. In certain applications such as in data centers, the inrush current during transformer energization can be significantly higher than the system's rated current, potentially reaching levels that trigger the protection circuit breakers to trip open to isolate the electrical load from the connected power source. Such high inrush currents also put significant stress on the upstream uninterruptible power supply (UPS) and may lead to malfunctions.

In some embodiments, a transfer switch includes a first switch including at least one first solid state switching device (SSSD) pair, the first switch electrically connecting a first power source to a load, a second switch including at least one second SSSD pair, the second switch electrically connecting a second power source to the load, the at least one first SSSD pair including a different type of switching device relative to the at least one second SSSD pair, and a controller is configured to selectively cycle the at least one first SSSD pair and the at least one second SSSD pair on and off to electrically couple one of the first power source and the second power source to the load.

In some embodiments, the first switch includes a plurality of first SSSD pairs, and each of the first SSSD pairs are connected in anti-series.

In some embodiments, each switching device of the at least one first SSSD pair is selected from a group including SiC metal-oxide-semiconductor field-effect transistors (MOSFETs), insulated gate bipolar transistors (IGBTs), bipolar junction transistors (BJTs), junction-gate field effect transistors (JFETs), or any combinations thereof.

In some embodiments, each switching device of the at least one first SSSD pair includes SiC MOSFETS.

In some embodiments, each switching device of the at least one first SSSD pair is selected from a group consisting of SiC MOSFETs, IGBTs, BJTs, JFETs, or any combinations thereof.

In some embodiments, the second switch includes a plurality of second SSSD pairs, the plurality of second SSSD pairs are connected in anti-parallel.

In some embodiments, each switching device of the at least one second SSSD pair is selected from a group including SiC MOSFETs, IGBTs, BJTs, JFETs, silicon controlled rectifiers (SCRs), SCRs with paralleled resistive turn-off (RTO) circuits, or any combinations thereof.

In some embodiments, each switching device of the at least one second SSSD pair includes SCRs with paralleled RTO circuits.

In some embodiments, each switching device of the at least one second SSSD pair is selected from a group consisting of SCRs, SCRs with paralleled RTO circuits, or any combinations thereof.

In some embodiments, the transfer switch further includes the controller, the controller including a processor and a non-transitory computer readable medium having stored thereon instructions executable by the processor to enable the controller to perform operations including send a first set of gate signals to the at least one first SSSD pair to open the first switch in response to detecting an abnormal condition at the first power source, and after a time delay, sending a second set of gate signals to the at least one second SSSD pair to close the second switch and to connect the load to the second power source.

In some embodiments, the time delay for transferring the load between the first power source and the second power source is less than 10 msec.

In some embodiments, the controller transfers the load from the first power source to the second power source when the first power source and the second power source are out of phase by more than 30°.

In some embodiments, a computer-implemented method for operating a transfer switch including a first switch and a second switch, the method including determining, by a controller, an abnormal condition at a first power source in electrical connection with the transfer switch, sending, by the controller and based on a first logic, a first set of gate signals to the first switch to open at least one first SSSD pair in response to detecting the abnormal condition at the first power source, and following a time delay, sending, by the controller and based on a second logic, a second set of gate signals to the second switch to close at least one second SSSD pair and to connect a load to a second power source, the at least one first SSSD pair includes a different type of switching devices relative to the at least one second SSSD pair.

In some embodiments, the delay time for transferring the load between the first power source and the second power source is less than 10 msec.

In some embodiments, the delay time for transferring the load is further configured to include when the first power source and the second power source are out of phase by more than 30°.

In some embodiments, the first logic is configured to operate switching devices of the at least one first SSSD pair, and the second logic is configured to operate switching devices of the at least one second SSSD pair.

In some embodiments, the first switch includes a plurality of first SSSD pairs, and the second switch includes a plurality of second SSSD pairs, each switching device in the plurality of first SSSD pairs are connected in anti-series and each switching device in the plurality of second SSSD pairs are connected in anti-parallel.

In some embodiments, a system including a first power source, a second power source, an electrical load, a transfer switch including a first switch including a plurality of first SSSD pairs connected in parallel between the first power source and the electrical load, and a second switch including a plurality of second SSSD pairs connected in parallel between the second power source and the electrical load, the plurality of first SSSD pairs includes different type of switching devices relative to the plurality of second SSSD pairs, and a controller including a processor, and a non-transitory computer readable medium having stored thereon instructions executable by the processor to enable the controller to perform operations including determine an abnormal condition at the first power source in electrical connection with the transfer switch, send a first set of gate signals to the plurality of first SSSD pairs to open the first switch in response to detecting an abnormal condition at the first power source, and after a time delay, sending a second set of gate signals to the plurality of second SSSD pairs to close the second switch and to connect the electrical load to the second power source.

In some embodiments, each switching device of the plurality of first SSSD pairs includes SiC MOSFETs connected in anti-series, each switching device of the plurality of second SSSD pairs includes SCRs with paralleled RTO circuits connected in anti-parallel.

In some embodiments, the time delay for transferring the electrical load between the first power source and the second power source is less than 10 msec, the controller transfers the electrical load from the first power source to the second power source when the first power source and the second power source are out of phase by more than 30°.

BRIEF DESCRIPTION OF THE DRAWINGS

Some embodiments of the disclosure are herein described, by way of example only, with reference to the accompanying drawings. With specific reference now to the drawings in detail, it is stressed that the embodiments shown are by way of example and for purposes of illustrative discussion of embodiments of the disclosure. In this regard, the description taken with the drawings makes apparent to those skilled in the art how embodiments of the disclosure may be practiced.

FIG. **1** illustrates a schematic diagram of a system, according to some embodiments.

FIG. **2** illustrates a schematic view of a portion of the system in FIG. **1**, according to some embodiments.

FIG. **3** illustrates a schematic view of a portion of the system in FIG. **1**, according to some embodiments.

FIG. **4** illustrates a schematic diagram of a portion of system, according to some embodiments.

FIG. **5** illustrates a schematic diagram of a portion of system, according to some embodiments.

FIG. **6** illustrates a schematic diagram of a non-limiting example of a RTO circuit, according to some embodiments.

FIG. **7** is a flow diagram of a method, according to some embodiments.

DETAILED DESCRIPTION

Conventional approaches to reducing inrush current include utilizing a high-grade steel core with a higher saturation flux density. Another approach enhances transformer core construction. Furthermore, a different approach incorporates additional impedance to the transformer windings. However, these conventional approaches to reduce inrush current come at the expense of increased costs to build the transformer core, which can result in requiring more complex transformer designs, and in the case of increasing the impedance in the windings, can lead to higher transformer losses.

In addition, conventional transfer switches can be constructed to mitigate inrush current. Conventional transfer switches typically include a first switch connecting a primary power source to the electrical load and a second switch connecting a secondary power source to the electrical load. When fluctuations or instability are detected at the primary power source, the transfer switch is triggered to switch from the first switch to the second switch. In conventional transfer switch topologies, the first switch and the second switch use the same type of semiconductor switching device, which may be a resonant turn off (RTO) thyristor based switch or a fully controlled power semiconductor switching device. Typically, the same type of switching device is used on both legs as different types of switching devices operate based on a different logics and typically requires a separate controller to operate each type of switching device. However, transfer switching including fully controlled power semiconductor devices may be more costly to manufacture and implement than using RTO thyristor based switching devices, whereas RTO thyristor based switching devices may be incapable of meeting the high switching frequencies demanded by certain applications.

Various embodiments of the present disclosure relate to systems, apparatus, and methods including a hybrid transfer switch that includes a first switch at one side or leg (e.g., connected to the primary power source) and a second switch at the other side or leg (e.g., connected to the secondary power source), the first switch and the second switch including different types of solid-state switching devices (SSSDs) therein relative to the other of the first switch and the second switch. The first switch may include one or more first SSSDs and the second switch may include one or more second SSSDs. In some embodiments, the first SSSDs may be fully controlled power semiconductor devices, and the second SSSDs may be RTO thyristor based switching devices. Moreover, in one or more embodiments, a controller may control the switching operations of the first switch and second switch and may include one or more logics configured to apply one or more algorithms to operate the different types of SSSDs in the first switch and the second switch, respectively, using a single controller. This increases the overall reliability of the system during an overload condition, thereby preventing interruptions in the power supply to the electrical load (e.g., data centers).

The controller may include therein a single logic circuit configured to operate switching devices of different types, thereby removing the need to include individual controllers to control different switching device types. In some embodiments, the single controller may operate the first switch including therein one or more first switching devices arranged in one or more switching pairs and may simultaneously operate the second switch including therein one or more second switching devices arranged in one or more switching pairs.

Various embodiments of the present disclosure may relate to a system including a transfer switch and a controller that controls an operation of the transfer switch including different types of SSSD in connection with the primary power source and the secondary power source, respectively. The controller includes a logic that is capable of controlling the switching operations of the two different types of SSSD to reduce inrush current and can prevent transformer failure due to excessive inrush current to upstream devices. Additionally, the various embodiments described herein can prevent the upstream breakers from tripping, thereby preventing a load drop for the electrical load. Accordingly, the various embodiments increase the life expectancy for upstream data center devices and enables the system to utilize transfer switches having a hybrid bridge topology, which reduces the cost of the system.

Among those benefits and improvements that have been disclosed, other objects and advantages of this disclosure will become apparent from the following description taken in conjunction with the accompanying figures. Detailed embodiments of the present disclosure are disclosed herein; however, it is to be understood that the disclosed embodiments are merely illustrative of the disclosure that may be embodied in various forms. In addition, each of the examples given regarding the various embodiments of the disclosure which are intended to be illustrative, and not restrictive.

FIG. **1** illustrates a schematic diagram of a system **100**, according to some embodiments.

The system **100** includes a transfer switch **102** which is conductively coupled to a first electrical power source **104**, hereinafter referred to as first source **104**, via a first mechanical circuit breaker (MCB) **106**, and a second electrical power source **108**, hereinafter referred to as second source **108**, which is conductively coupled with the transfer switch **102** via a second MCB **110**. In some embodiments, the system **100** may be electrically connected to a transformer **112** at a primary side which is conductively coupled with the transfer switch **102** via a third MCB **114** and a fourth MCB **116**, and a secondary side which is conductively coupled with a load **120**.

The first source **104** and the second source **108** may be a number of forms and types of electrical power sources, for example, a utility grid, a microgrid, a nanogrid, a backup generator, an uninterruptable power supply (UPS) or backup battery, a flywheel operatively coupled with a motor/generator, a PV array, a wind farm, a fuel cell installation, or any of a number of other sources of electrical power as will occur to one of skill in the art with the benefit of the present disclosure. One of the first source **104** and the second source **108** may be a primary or preferred power source for the system **100** and the other of the first source **104** and the second source **108** may be a secondary or backup power source for the system **100**. In some embodiments, the first source **104** may be a utility grid serving as a primary power source and the second source **108** may be one or more UPS serving as a backup power source. In some embodiments, the transfer switch **102** may also be considered and referred to as a bypass switch or a UPS bypass switch. The load **120** may be any of a variety of types of load systems, for example, a datacenter, educational facility, governmental facility, hospital or other healthcare facility, manufacturing, chemical or other industrial plant, water treatment plant, or other types of loads or load systems as will occur to one of skill in the art with the benefit of the present disclosure.

The MCB **106**, **110**, **114**, **116** are configured and operable to provide fault protection by transitioning from a closed-circuit state to an open-circuit state in response to a fault condition, such as an over-current condition, an over-voltage condition, and/or another fault condition. Furthermore, the MCB **106**, **110**, **114**, **116** may be configured and operable to provide passive fault protection or other active opening or closing operation (e.g., in response to control signals received from the electronic control system (ECS) **122**), or both. It shall be appreciated that certain embodiments may omit one or more of the MCB **106**, **110**, **114**, **116**. Furthermore, certain embodiments may comprise additional or alternate fault protection devices as will occur to one of skill in the art with the benefit of the present disclosure.

The system **100** may also include a fifth MCB **124**, which may be conductively coupled between the first source **104** and the transformer **112**, and a sixth MCB **126**, which may be conductively coupled between the second source **108** and the transformer **112**, according to some embodiments. The MCB **124**, **126** are configured to selectably provide a closed circuit connection between the first source **104** and the second source **108**, respectively, bypassing the transfer switch **102** and may be actively controlled by the ECS **122**, hereinafter referred to as controller **122**. It shall be appreciated that certain embodiments may omit one or both of the MCB **124**, **126**. Furthermore, certain embodiments may include additional or alternate bypass devices as will occur to one of skill in the art with the benefit of the present disclosure.

The transfer switch **102** includes switch **130** and switch **132**. Switch **130** and switch **132** can be controlled by controller **122** to energize (or deenergize) the transformer **112** by conductively coupling (or decoupling) the transformer **112** with the first source **104** or the second source **108**, respectively. Switch **130** includes one or more SSSDs **134** arranged in switching pairs to enable a bi-directional flow of current at switch **130**. Switch **132** includes one or more SSSDs **136** arranged in switching pairs to enable a bi-directional flow of current at switch **132**. The SSSDs **134**, **136** arranged in switch **130** and switch **132** may include, but are not limited to, SiC metal-oxide-semiconductor field-effect transistors (MOSFETs), insulated gate bipolar transistors (IGBTs), bipolar junction transistors (BJTs), junction-gate field effect transistors (JFETs), silicon controlled rectifiers (SCRs), SCRs with paralleled resistive turn-off (RTO) circuits, or any combinations thereof, according to some embodiments.

In addition, the SSSDs **134** in switch **130** may be one type of switching device and the SSSDs **136** in switch **132** may be a different type of switching device distinct from the SSSDs **134** in switch **130**. For example, the SSSDs **134** in switch **130** may be SiC MOSFETs, and the SSSDs **136** in switch **132** may be SCRs. In some embodiments, the SSSDs **134** in switch **130** may be fully controlled power semiconductor switching devices including the group of SiC MOSFETS, IGBTs, BJTs, JFETs, other fully controlled power semiconductor switching devices, or any combinations thereof. In other embodiments, the SSSDs **134** in switch **130** may be fully controlled power semiconductor switching devices consisting of the group of SiC MOSFETS, IGBTs, BJTs, JFETs, or any combinations thereof. Moreover, in some embodiments, the SSSDs **136** in switch **132** may be thyristor based switching devices including silicon controlled rectifiers (SCRs), SCRs with paralleled resistive turn-off (RTO) circuits, or any combinations thereof. In other embodiments, the SSSDs **136** in switch **132** may be thyristor based switching devices consisting of the group of silicon controlled rectifiers (SCRs), SCRs with paralleled resistive turn-off (RTO) circuits, or any combinations thereof.

Controller **122** may selectively send gate control signals to switch **130** and switch **132** to control the cycling of the respective switch, and the respective SSSDs therein, on and off and to connect/disconnect the load **120** from the first source **104** and/or the second source **108**, respectively. Controller **122** may control the operation of switch **130** to cycle to an on or closed state to conductively couple the first source **104** with the transformer **112** and load **120** (provided that the MCB **106**, **114** are in a closed state) and to an off or open state which conductively decouples the first source **104** with the transformer **112**. Switch **132** may be controlled by controller **122** to an on or closed state which conductively couples the second source **108** with the transformer **112** (provided that the MCB **110**, **116** are in a closed state) and to an off or open state which conductively decouples the second source **108** with the transformer **112**. The switches **130** may be provided in a number of configurations and forms including, for example, the forms illustrated and described below in connection with FIGS. **2** and **3**.

The controller **122** may include a processor and a memory. The memory may be a computer readable medium having stored thereon instructions executable by the processor to perform operations in accordance with the present disclosure. The controller **122** includes a logic capable of controlling an operation of the transfer switch **102** including sending one or more gating signals to control an operation of switches **130**. The controller **122** may include one or more other components, including components capable of placing the controller **122** in electrically communicable connection with the transfer switch **102** and to monitor and control the operation of the transfer switch **102** based on one or more parameters measured at the transfer switch **102** such as, for example, by one or more sensors (not shown). In some embodiments, the controller **122** may be operatively coupled with the transfer switch **102**.

Additionally, in some embodiments, the controller **122** may in some forms also be operatively coupled with one or more of the MCB **106**, **110**, **114**, **116**, **124**, **126** and may monitor and/or actively control one or more of the MCB **106**, **110**, **114**, **116**, **124**, **126**. The controller **122** may be provided as a portion or component of the transfer switch **102** (e.g., provided in a common housing or as a common unit), as one or more separate components, or distributed among one or more components forming a portion of the transfer switch **102** and one or more separate components. The controller **122** may include one or more integrated circuit-based (e.g., microprocessor-based, microcontroller-based, ASIC-based, FPGA-based, and/or DSP-based) control units as well as related driver, input/output, signal conditioning, signal conversion, non-transitory machine-readable memory devices storing executable instructions, and other circuitry.

It shall be appreciated that system **100** may be provided in a single-phase form, a three-phase form, or other multi-phase forms. Such multi-phase forms, the first source **104** and the second source **108** may be multi-phase power sources (e.g., three-phase power sources). In such forms, the MCB **106**, **110**, **114**, **116**, **124**, **126**, the transfer switch **102** and its constituent switches including switch **130** and switch **132**, and the transformer **112**, may be provided in corresponding multi-phase forms and arrangements (e.g., three-phase forms and arrangements) wherein an additional instance of these components may be provided to service each additional phase. Furthermore, while system **100** is illustrated as comprising a first source **104** and a second source **108**, it shall be appreciated that additional sources may also be present in certain embodiments and that such additional sources may include additional respective MCB components for fault protection and bypass operation and additional respective constituent switch **130** and/or switch **132** of the transfer switch **102**.

In some embodiments, the switch **130** may include at least one switching pair of SSSDs **134** consisting of SiC MOSFETs, and switch **132** may include at least one switching pair of SSSDs **136** consisting of SCRs. The switch **130** including the SiC MOSFETs may be connected to the first source **104** due to the SSSDs **134** fast switching capability in case of an abnormal condition at the first source **104**, which reduces the transfer time for transferring power from the first source **104** to the second source **108** in case of an abnormal condition. For example, the SiC MOSFET may have a transfer time of 10 msec, whereas the SCR may have a transfer time of more than 16 msec. However, as SiC MOSFET's have low surge current capability, transferring to the switch **132** side with the SCRs is essential during overload conditions.

Conventional transfer switch topologies typically utilize the same type of switching device at each leg of the transfer switch. As such, both the first switch and second switch that connects the load to the first power source and the second power source, respectively, include the same type of switching device. Conventional transfer switch topologies typically do not utilize hybrid topologies as different types of switching devices operate based on different logics. Accordingly, conventional systems of the prior art do not utilize such hybrid transfer switch topologies as doing so requires a controller for each switch to operate the different types of respective switching devices therein based on the different logics. Furthermore, conventional transfer switches of the prior art typically utilize switches including fully controlled power semiconductor devices (e.g., SiC MOSFETs) therein over SCR based thyristors arranged in a parallel configuration between the load and the power source as such switches connected to the primary power source can be difficult to operate and such SCR based switches have certain requirements for transferring power that can cause delays in switching, thereby leading to an increased likelihood of causing damage to downstream components and loads.

FIG. **2** illustrates a schematic view of a portion of the system **100** in FIG. **1**, according to some embodiments.

The system **200** includes a modular static transfer switch (MSTS) **202** that selectively supplies a load **220** with electrical power from either a first power source **204** or a second power source **208** depending on various criteria. For example, MSTS **202** may supply electrical power to load **220** primarily from first power source **204** unless the electrical power delivered by first power source **204** falls outside of a desired range of values (e.g., first power source **204** has a voltage and/or a harmonic distortion that varies from target values by a threshold amount). If, for example, first power source **204** is either incapable of supplying electrical power to load **220** (e.g., first power source **204** fails or is incapable of supplying electrical power to load **220** at a desired power quality), then MSTS **202** switches load **220** from first power source **204** to second power source **208**. In this regard, first power source **204** may operate as a preferred power source for load **220**, with second power source **208** operating as a backup or alternate power source for load **220**. Although only two power sources are depicted in FIG. **2**, MSTS **202** selectively couples load **220** to any number of power sources in other embodiments. Further, although MSTS **202** is depicted as switching single phase power in FIG. **2**, MSTS **202** switches 3-phase Alternating Current (AC) power in other embodiments. In 3-phase embodiments, first power source **204**, second power source **208** are 3-phase sources, and load **220** is a 3-phase load. In other embodiments, first power source **204** and second power source **208** are Direct

Current (DC) sources, and load **220** is a DC load. In other embodiments, first power source **204** and second power source **208** are 3-phase sources, and MSTS provides a plurality of sign-phase loads (e.g., load **220** is a plurality of single-phase loads).

The first power source **204** is electrically coupled to MSTS **202** at a first input **210** and second power source **208** is electrically coupled to MSTS **202** at a second input **212**. Load **220** is electrically coupled to an output **218** of MSTS **202**. First input **210** is electrically coupled to a first input bus **224** of MSTS **202** and second input **212** is electrically coupled to a second input bus **226** of MSTS **202**. Output **218** of MSTS **202** is electrically coupled to an output bus **228** of MSTS **202**.

MSTS **202** includes a plurality of solid-state switch modules **230** and a plurality of solid-state switch modules **232**. Switch modules **230** may correspond to the switch **130** and switch modules **232** may correspond to the switch **132**, as shown in FIG. **1**, according to some embodiments. The switch modules **230** operate to selectively couple first power source **204** to load **220** and the switch modules **232** operate to selectively couple the second power source **206** to load **220**. It is to be appreciated by those having ordinary skill in the art that modules **230** and modules **232** may further include any components, systems, or devices which enables the modules **230** and modules **232** to selectively couple load **220** to the first power source **204** or second power source **208**, respectively. In some embodiments, modules **230** includes one or more SSSDs **234**, which provide a selective bi-directional electrical path between first power source **204** and load **220**, and modules **232** includes one or more SSSDs **236**, which provide a selective bi-directional electrical path between second power source **208** and load **220**. In some embodiments, the SSSDs **234** may correspond to the SSSDs **134** and the SSSDs **236** may correspond to the SSSDs **136** as shown in FIG. **1**. The one or more SSSDs **234** and the one or more SSSDs **236** may include any number of combinations of power electronic devices connected in series in order to implement the functionality described herein for switch modules **230** and switch modules **232**, respectively. For example, the one or more SSSDs **234** may be connected in anti-series.

Modules **230** are electrically coupled in parallel between first input bus **224** and output bus **228**. Modules **232** are electrically coupled in parallel between second input bus **226** and output bus **228**. Although only three modules **230** and only three modules **232** are depicted, the MSTS **202** may include any number of modules **230** and modules **232** electrically coupled in parallel in other embodiments. Further, modules **230** and modules **232** include 3-phase inputs and 3-phase outputs in other embodiments, depending on whether first power source **204**, second power source **208**, and load **220** are 3-phase sources/loads.

MSTS **202** further includes a back panel **246**, and modules **230** and modules **232** are removably mounted to back panel **246**. When removably mounted, modules **230** and modules **232** may be easily replaced during maintenance, even if MSTS **202** remains energized by first power source **204** or second power source **208** and is supplying electrical power to load **220**. Back panel **246** includes, in some embodiments, a communication bus **248**, which enables modules **230** to communicate with each other and enables modules **232** to communicate with each other. Communication bus **248** includes any component, system, or device that provides wired or wireless communication capability to a user, modules **230**, modules **232**, and MSTS **202**. During operation, modules **230** and modules **232** may utilize communication bus **248** to coordinate their activities during transfer events. For instance, modules **230** and modules **232** may communicate with each other utilizing communication bus **248** to collaboratively transfer load **220** between first power source **204** and second power source **208**.

MSTS **202** includes, in some embodiments, a management controller **222**. Management controller **222** includes any component, system, or device that provides management functions for MSTS **202**. According to some embodiments, the controller **222** may correspond to the controller **122** as shown in FIG. **1**. In some embodiments, management controller **222** communicates with modules **230** and modules **232** via communication bus **248** to manage transfers of load **220** between first power source **204** and second power source **208**. Management controller **222** therefore may perform any functionality described herein for MSTS **202** either alone or in combination with one or more of modules **230** and modules **232** in order to perform the functionality described herein for MSTS **202**.

FIG. **3** illustrates a schematic view of a portion of the system **100** in FIG. **1**, according to some embodiments.

MSTS **202** includes a non-limiting example of module **230**, according to some embodiments. Module **230** includes electrical terminals **252**, **254**, which are electrically couplable between output bus **228** in MSTS **202** and first input bus **224**. Module **230** in this embodiment further includes a controller **222**, a communication interface **258**, a sensing circuit **260**, a coil driver circuit **262**, a gate driver circuit **264**, a mechanical disconnect **266**, and a snubber circuit **268**. Controller **222**, communication interface **258**, sensing circuit **260**, coil driver circuit **262**, gate driver circuit **264**, mechanical disconnect **266**, and snubber circuit **268** include any component, system, or device which implements their respective functionality as described herein.

Controller **222** controls the operation of module **230** and the SSSDs **234** therein and interacts with communication interface **258** to send and/or receive information over communication bus **248**. Controller **222** utilizes gate driver circuit **264** to control whether SSSDs **234** are open or closed. Controller **222** also utilizes coil driver circuit **262** to control whether mechanical disconnect **266** is open or closed. Generally, mechanical disconnect **266** provides galvanic isolation for module **230** when mechanical disconnect **266** is open.

Sensing circuit **260** measures information for module **230**, including a temperature of module **230**, a humidity, a current flowing between terminals **252**, **254** (including a current flowing through SSSDs **234**), a voltage at terminals **252**, **254** and/or a voltage at SSSD **234**, a power factor at terminals **252**, **254**, a harmonic distortion at terminals **252**, **254**, etc.

Module **230** utilizes snubber circuit **268** during on-off transitions of SSSDs **234** to clamp voltages transients across SSSDs **234**. The components depicted for snubber circuit **268** are illustrative only, and snubber circuit **268** has different configurations in other embodiments. In this embodiment, SSSDs **234** is depicted as a pair of anti-series MOSFETs. In some embodiments, MOSFETs are arranged in serial and parallel combinations depending on the current capability of module **230**.

Although module **230** in this embodiment is depicted as a single-phase device, module **230** is a 3-phase device in other embodiments. In 3-phase embodiments, module **230** includes additional instances of terminals **252**, **254**, SSSD **134**, mechanical disconnect **266**, and snubber circuits **268** for each phase.

In some embodiments, controller **222** modifies the operation of module **230** based on information measured by sensing circuit **260** by opening and closing mechanical disconnect **266** and/or SSSDs **234** based on the temperature of module **230**, a humidity, the current flowing between terminals **252**, **254** (including a current flowing through SSSDs **234**), the voltage at terminals **252**, **254** and/or a voltage at SSSDs **234**, the power factor at terminals **252**, **254**, the harmonic distortion at terminals **252**, **254**, etc. For example, during current faults, currents higher than a threshold current may flow through SSSDs **234**, which are sensed by controller **222** and cause controller **222** to open SSSDs **234** and/or mechanical disconnect **266**. Such fault currents may further be detected by comparing the measured currents to a pre-determined time-current curve.

In another example, a temperature of module **230** higher than a threshold temperature may cause controller **222** to open SSSDs **234** and/or mechanical disconnect **266**. In another embodiment, the voltage at terminals **252**, **254** and/or a voltage at SSSDs **234**, the power factor at terminals **252**, **254**, the harmonic distortion at terminals **252**, **254**, a humidity, etc., may cause controller **222** to initiate a transfer between first power source **204** and second power source **208**. In other embodiments, any of the prior factors may trigger controller **222** to communicate this information to management controller **222**, which may take any further action deemed appropriate in order to rectify non-standard operating conditions at MSTS **202**.

In some embodiments, the modules **232** may include one or more of the circuits, components, or any combinations thereof that are included in modules **230** to enable the modules **232** to perform operations similar to modules **230** for connecting the load **220** to the second power source **208**. It is to be appreciated by those having ordinary skill in the art that the modules **230** and modules **232** may include one or more of the above described components and/or circuits described above. It is also to be appreciated by those having ordinary skill in the art that the modules **230** and modules **232** may include other components not described herein to enable the MSTS **202** to perform the operations in accordance with the present disclosure.

FIG. **4** illustrates a schematic diagram of a portion of system **100**, according to some embodiments.

Switch **130** includes one or more SSSDs **134** arranged in switching pairs connected in an anti-series arrangement to enable a bi-directional flow of current at switch **130**. Although switch **130** in FIG. **4** only shows one leg including SSSD **134**_a_ and SSSD **134**_b_ arranged in anti-series, it is to be appreciated that the switch **130** may include one or more legs connected in parallel between the first source **104** and transformer **112** and/or load **120**. As such, each leg of switch **130** may include SSSD **134**_a_ and SSSD **134**_b_ arranged in anti-series.

The SSSDs **134** arranged in switch **130** may include, but are not limited to, SiC metal-oxide-semiconductor field-effect transistors (MOSFETs), insulated gate bipolar transistors (IGBTs), bipolar junction transistors (BJTs), junction-gate field effect transistors (JFETs), silicon controlled rectifiers (SCRs), SCRs with paralleled resistive turn-off (RTO) circuits, or any combinations thereof, according to some embodiments. In some embodiments, the SSSDs **134** may be selected from a group including SiC MOSFETS, IGBTs, BJTs, JFETs, other fully controlled switching devices, or any combinations thereof. In other embodiments, the SSSDs **134** may be selected from a group consisting of SiC MOSFETS, IGBTs, BJTs, JFETs, other fully controlled switching devices, or any combinations thereof.

FIG. **5** illustrates a schematic diagram of a portion of system **100**, according to some embodiments.

Switch **132** includes one or more SSSDs **136** arranged in switching pairs connected in an anti-parallel arrangement to enable switching at positive and negative half cycles of current at switch **132**. Although switch **132** in FIG. **5** only shows one leg including SSSD **136**_a_ and SSSD **136**_b_ arranged in anti-parallel, it is to be appreciated that the switch **132** may include one or more legs connected in parallel between the first source **108** and transformer **112** and/or load **120**. As such, each leg of switch **132** may include SSSD **136**_a_ and SSSD **136**_b_ arranged in anti-parallel.

The SSSDs **136** arranged in switch **132** may include, but are not limited to, SiC metal-oxide-semiconductor field-effect transistors (MOSFETs), insulated gate bipolar transistors (IGBTs), bipolar junction transistors (BJTs), junction-gate field effect transistors (JFETs), silicon controlled rectifiers (SCRs), SCRs with paralleled resistive turn-off (RTO) circuits **138**, or any combinations thereof, according to some embodiments. In some embodiments, the SSSDs **136** may be selected from a group including SiC MOSFETs, IGBTs, BJTs, JFETs, SCRs, SCRs with paralleled RTO circuits **138**, other switching devices, or any combinations thereof. In other embodiments, the SSSDs **136** may be selected from a group including SCRs, SCRs with paralleled RTO circuits **138**, other thyristor based switching devices, or any combinations thereof. In yet other embodiments, the SSSDs **136** may be selected from a group consisting of SCRs, SCRs with paralleled RTO circuits **138**, other thyristor based switching devices, or any combinations thereof.

FIG. **6** illustrates a schematic diagram of a non-limiting example of an RTO circuit **140**, according to some embodiments.

The RTO circuit **140** is connected in parallel with the SSSD **136** pair in switch **132** and structured to receive power and output a resonant current to the pair of SSSDs **136**. It shall be appreciated that RTO circuit **140** as shown in FIG. **5** is an example embodiment of a resonant circuit according to the present disclosure. Other embodiments may include a number of additions, modifications, or alternative resonant circuit arrangements including different types and arrangements of legs, switching devices, and capacitors.

The RTO circuit **140** includes four switching devices **146**, resonant capacitor **148** for energy storage, pre-charged capacitor **150**, and resonant inductor **152**. In some embodiments, the RTO circuit **140** includes a voltage clamp **156**. The voltage clamp **156** may be a metal oxide varistor (MOV), in some embodiments. Other embodiments may include other types or arrangements of voltage clamps. In the illustrated embodiment, the RTO switching devices **146** comprise thyristors. In other embodiments, the RTO switching devices may comprise other types of semiconductor switching devices such as insulated gate bipolar transistors (IGBT).

The RTO circuit **140** provides three possible design choices to control performance: resonant capacitance value C from resonant capacitor **148**, pre-charged initial capacitor voltage Vc**0** from pre-charged capacitor **150**, and resonant inductance value L from inductor **152**. These parameters can be used to determine how much and how fast the main thyristor current can be turned off, as well as the size and cost of the auxiliary resonant circuit. The pre-charged capacitor **150** may be pre-charged by pre-charge circuit **154** to provide resonant current to create a zero-current crossing for the SSSDs **136** in switch **132**. The inductor **152** limits di/dt for SSSDs **136** during turn-off. During normal conduction, only the SSSDs **136** are conducting and all the auxiliary switches **146** are off. Thus, the pre-charged resonant capacitor **150** is isolated from the SSSDs **136**. In resonant turn-off operation, the corresponding auxiliary switches **146** (depending on current direction) are triggered to open by sending a gate signal thereto. As a result, the energy stored in the resonant capacitor **148** is discharged. When the resonant current through the inductor **152** exceeds the load current, the current through the SSSDs **136** is commuted to the resonant circuit. In the meantime, the capacitor **148** voltage provides a negative bias voltage to help the respective SSSDs **136** to turn off (i.e., open and stop conducting). When the respective SSSDs **136** current reaches zero, it starts turning off with reverse bias voltage from the resonant capacitor **148**. All prior patents and publications referenced herein are incorporated by reference in their entireties.

The switches **132** is structured to open in response to control signals or commands received from the controller **122**. To reduce the time necessary for opening the switches **132**, the RTO circuit **140** may be configured and operable to output resonant current to the SSSDs **136** to force commutation and to thereby turn off. The RTO circuit **140** may thereby increase the speed at which the SSSDs **136** operate to open the switch **132** (e.g., 80% faster compared to the same SSSDs **136** without the RTO circuit **140**). Further details of the operation and control of the RTO circuit **140** and switch **132** (and switches therein) may be found in International Application No. PCT/US20/64217, filed Dec. 10, 2020, the disclosure of which is hereby incorporated by reference. It shall be appreciated that the SSSD **136** is one example of an SSSD which is quasi-fully controllable by assisted or forced commutation allowing a thyristor or other semi-controllable device, which can be turned off under only certain conditions (e.g., zero current conditions) to function as a fully controlled device via the assisted or forced commutation.

The controller **122** may be operatively coupled to the SSSDs **136** in the SSSDs **136** and the controller **122** may selectively send a set of gate signals to SSSDs **136** to selectably turn each of the SSSDs **136** on (e.g., a closed or conductive state) or off (e.g., an open or non-conductive state). The controller **122** may also be operatively coupled with the RTO circuit **140** and configured to provide control signals to the four switching devices **146** to selectably turn each of these switching devices on (e.g., a closed or conductive state) or off (e.g., an open or non-conductive state). The controller **122** may also be configured and operable to receive one or more inputs indicative of voltage, current, and/or flux values at one or more nodes of system **100** and to control system **100** as further described herein.

The controller **122** operates the RTO circuit **140** to generate and provide a resonant current (IR) configured and operable to force commutation of the SSSDs **136**. The resonant current (IR) causes the magnitude of the current conducted by SSSDs **136** to decrease to zero and causes a reverse voltage bias across the SSSDs **136**.

It shall be appreciated that the RTO circuit **140** is one example embodiment of a resonant circuit that may be coupled in parallel with a thyristor-based SSSD such as SSSDs **136** and utilized to increase a thyristor turn-off speed by injecting a reverse current to force the thyristor current to commutate to zero. Using RTO circuit or other forms of parallel resonant circuits, thyristor-based forms the switching pair **136** (and SSSDs therein) can interrupt the current.

FIG. **7** is a flow diagram of a method **300**, according to some embodiments.

At **302**, the method **300** includes determining an abnormal condition at a first power source **104** in electrical connection with the transfer switch **102**. The transfer switch **102** may connect the first source **104** and the second source **108** to the load **120** and/or the transformer **112**. The transfer switch **102** includes switch **130** connected between first source **104** and load **120** and includes switch **132** connected between second source **108** and load **120**. In some embodiments, switch **130** includes a pair of first SSSDs **134** and switch **132** includes a pair of second SSSDs **136**. In other embodiments, switch **130** includes a plurality of pairs of first SSSDs **134** arranged in parallel between the first source **104** and load **120** and switch **132** includes a plurality of pairs of SSSDs **136** arranged in parallel between second source **108** and load **120**. In addition, at each of pair of SSSDs **134**, the SSSDs **134** may be arranged in anti-series configuration, and at each pair of SSSDs **136**, the SSSDs **136** may be arranged in anti-parallel configuration.

At **304**, the method **300** includes sending, based on a first logic, a first set of gate signals to the first switch **130** to open at least one first SSSD **134** pair in response to detecting the abnormal condition at the first power source **104** to disconnect the load **120** from the first source **104** to protect the downstream electrical components. At **306**, the method **300** includes following a time delay, sending, based on a second logic, a second set of gate signals to the second switch **132** to close at least one second SSSD **136** pair and to connect a load **120** to a second power source **108** to maintain uninterrupted power at the load **120**. In some embodiments, the first SSSD **134** pairs in the switch **130** comprises a different type of switching devices relative to the at least one second SSSD **136** pairs in the switch **132**. Moreover, in some embodiments, the delay time for transferring the load **120** between the first power source **104** and the second power source **108** based on an operation of the switch **130** may be less than 10 msec. Further, in some embodiments, the delay time for transferring the load **120** may be further configured to include when the first power source **104** and the second power source **108** are out of phase by more than 30°.

Controller **122** may monitor the condition of the first source **104** and second source **108** and may also determine whether an abnormal condition is present at the first source **104** or second source **108**. Further, based on determining the presence of an abnormal condition at one of the first source **104** or second source **108**, the controller **122** may control the operation of the transfer switch **102** to cause the switch **130** and switch **132** to selectively cycle on/off to transfer the load **120** from one of the first source **104** and second source **108** to the other of the first source **104** and second source **108**.

To transfer the load **120** from the first source **104** to second source **108**, or vice versa, controller **122** may generate the first set of signals to selectively cycle the first pair of SSSDs **134** and the second set of signals to selectively cycle the second pair of SSSDs **136** to facilitate transferring the load **120** to one of the first source **104** or second source **108**. As the SSSDs **134** in switch **130** are of a different type from the SSSDs **136** in switch **132**, the controller **122** may include therein a first logic to operate the SSSDs **134** and a second logic to operate the SSSDs **136**. In addition, the controller **122** may include therein a logic circuit that includes both the first logic and the second logic such that the controller **122** may control the SSSDs **134** in switch **130** and SSSDs **136** in switch **132** without needing additional controllers or additional logic circuits therein. The first logic and the second logic of controller **122** may be configured to apply one or more algorithmic functions to enable the controller **122** to selectively control the cycling of the SSSDs **134** in switch **130** and the cycling of the SSSDs **136** in switch **132** to facilitate switching the load **120** from the first source **104** to the second source **108** or vice versa.

All prior patents and publications referenced herein are incorporated by reference in their entireties.

Throughout the specification and claims, the following terms take the meanings explicitly associated herein, unless the context clearly dictates otherwise. The phrases "in one embodiment," "in an embodiment," and "in some embodiments" as used herein do not necessarily refer to the same embodiment(s), though it may. Furthermore, the phrases "in another embodiment" and "in some other embodiments" as used herein do not necessarily refer to a different embodiment, although it may. All embodiments of the disclosure are intended to be combinable without departing from the scope or spirit of the disclosure.

As used herein, the term "based on" is not exclusive and allows for being based on additional factors not described, unless the context clearly dictates otherwise. In addition, throughout the specification, the meaning of "a," "an," and "the" include plural references. The meaning of "in" includes "in" and "on."

As used herein, the term "between" does not necessarily require being disposed directly next to other elements. Generally, this term means a configuration where something is sandwiched by two or more other things. At the same time, the term "between" can describe something that is directly next to two opposing things. Accordingly, in any one or more of the embodiments disclosed herein, a particular structural component being disposed between two other structural elements can be:

- disposed directly between both of the two other structural elements such that the particular structural component is in direct contact with both of the two other structural elements;
- disposed directly next to only one of the two other structural elements such that the particular structural component is in direct contact with only one of the two other structural elements;
- disposed indirectly next to only one of the two other structural elements such that the particular structural component is not in direct contact with only one of the two other structural elements, and there is another element which juxtaposes the particular structural component and the one of the two other structural elements;
- disposed indirectly between both of the two other structural elements such that the particular structural component is not in direct contact with both of the two other structural elements, and other features can be disposed therebetween; or any combination(s) thereof.

As used herein "embedded" means that a first material is distributed throughout a second material.

Aspects

Various Aspects are described below. It is to be understood that any one or more of the features recited in the following Aspect(s) can be combined with any one or more other Aspect(s).

Aspect 1. A transfer switch comprising: a first switch comprising: at least one first solid state switching device (SSSD) pair, wherein the first switch electrically connects a first power source to a load; a second switch comprising: at least one second SSSD pair, wherein the second switch electrically connects a second power source to the load; and wherein the at least one first SSSD pair comprises a different type of switching devices relative to the at least one second SSSD pair; wherein a controller is configured to selectively cycle the at least one first SSSD pair and the at least one second SSSD pair on and off to electrically couple one of the first power source and the second power source to the load.

Aspect 2. The transfer switch according to aspect 1, wherein the first switch comprises a plurality of first SSSD pairs, wherein each of the first SSSD pairs are connected in anti-series.

Aspect 3. The transfer switch according to any of the preceding aspects, wherein each switching device of the at least one first SSSD pair is selected from a group comprising SiC metal-oxide-semiconductor field-effect transistors (MOSFETs), insulated gate bipolar transistors (IGBTs), bipolar junction transistors (BJTs), junction-gate field effect transistors (JFETs), or any combinations thereof.

Aspect 4. The transfer switch according to aspect 3, wherein each switching device of the at least one first SSSD pair comprises SiC MOSFETs.

Aspect 5. The transfer switch according to any of the preceding aspects, wherein each switching device of the at least one first SSSD pair is selected from a group consisting of SiC MOSFETS, IGBTs, BJTs, JFETs, or any combinations thereof.

Aspect 6. The transfer switch according to any of the preceding aspects, wherein the second switch comprises a plurality of second SSSD pairs, wherein the plurality of second SSSD pairs are connected in anti-parallel.

Aspect 7. The transfer switch according to any of the preceding aspects, wherein each switching device of the at least one second SSSD pair is selected from a group comprising SiC MOSFETs, IGBTs, BJTS, JFETs, silicon controlled rectifiers (SCRs), SCRs with paralleled resistive turn-off (RTO) circuits, or any combinations thereof.

Aspect 8. The transfer switch according to any of the preceding aspects, wherein each switching device of the at least one second SSSD pair comprises: SCRs with paralleled RTO circuits.

Aspect 9. The transfer switch according to any of the preceding aspects, wherein each switching device of the at least one second SSSD pair is selected from a group consisting of SCRs, SCRs with paralleled RTO circuits, or any combinations thereof.

Aspect 10. The transfer switch according to any of the preceding aspects, further comprising: the controller, the controller comprises a processor and a non-transitory computer readable medium having stored thereon instructions executable by the processor to enable the controller to perform operations comprising: send a first set of gate signals to the at least one first SSSD pair to open the first switch in response to detecting an abnormal condition at the first power source, and after a time delay, sending a second set of gate signals to the at least one second SSSD pair to close the second switch and to connect the load to the second power source.

Aspect 11. The transfer switch according to aspect 10, wherein the time delay for transferring the load between the first power source and the second power source is less than 10 msec.

Aspect 12. The transfer switch according to aspects 10 or 11, wherein the controller transfers the load from the first power source to the second power source when the first power source and the second power source are out of phase by more than 30°.

Aspect 13. A computer-implemented method for operating a transfer switch including a first switch and a second switch, the method comprising: determining, by a controller, an abnormal condition at a first power source in electrical connection with the transfer switch; sending, by the controller and based on a first logic, a first set of gate signals to the first switch to open at least one first SSSD pair in response to detecting the abnormal condition at the first power source; and following a time delay, sending, by the controller and based on a second logic, a second set of gate signals to the second switch to close at least one second SSSD pair and to connect a load to a second power source, wherein the at least one first SSSD pair comprises a different type of switching devices relative to the at least one second SSSD pair.

Aspect 14. The computer-implemented method according to aspect 13, wherein the delay time for transferring the load between the first power source and the second power source is less than 10 msec.

Aspect 15. The computer-implemented method according to aspects 13 or 14, wherein the delay time for transferring the load is further configured to include when the first power source and the second power source are out of phase by more than 30°.

Aspect 16. The computer-implemented method according to aspects 13, 14, or 15, wherein the first logic is configured to operate switching devices of the at least one first SSSD pair, and the second logic is configured to operate switching devices of the at least one second SSSD pair.

Aspect 17. The method according to aspects 13, 14, 15, or 16, wherein the first switch comprises a plurality of first SSSD pairs, and the second switch comprises a plurality of second SSSD pairs, wherein each switching device in the plurality of first SSSD pairs are connected in anti-series and each switching device in the plurality of second SSSD pairs are connected in anti-parallel.

Aspect 18. A system comprising: a first power source; a second power source; an electrical load; a transfer switch comprising: a first switch comprising a plurality of first SSSD pairs connected in parallel between the first power source and the electrical load, and a second switch comprising a plurality of second SSSD pairs connected in parallel between the second power source and the electrical load, wherein the plurality of first SSSD pairs comprises different type of switching devices relative to the plurality of second SSSD pairs; and a controller comprising: a processor, and a non-transitory computer readable medium having stored thereon instructions executable by the processor to enable the controller to perform operations comprising: determine an abnormal condition at the first power source in electrical connection with the transfer switch, send a first set of gate signals to the plurality of first SSSD pairs to open the first switch in response to detecting an abnormal condition at the first power source, and after a time delay, sending a second set of gate signals to the plurality of second SSSD pairs to close the second switch and to connect the electrical load to the second power source.

Aspect 19. The system according to aspect 18, wherein each switching device of the plurality of first SSSD pairs comprises SiC MOSFETs connected in anti-series, wherein each switching device of the plurality of second SSSD pairs comprises SCRs with paralleled RTO circuits connected in anti-parallel.

Aspect 20. The system according to aspects 18 or 19, wherein the time delay for transferring the electrical load between the first power source and the second power source is less than 10 msec, wherein the controller transfers the electrical load from the first power source to the second power source when the first power source and the second power source are out of phase by more than 30°.

It is to be understood that changes may be made in detail, especially in matters of the construction materials employed and the shape, size, and arrangement of parts without departing from the scope of the present disclosure. This Specification and the embodiments described are examples, with the true scope and spirit of the disclosure being indicated by the claims that follow.

What is claimed is:

1. A transfer switch comprising:

a first switch comprising:

at least one first solid state switching device (SSSD) pair,

wherein the first switch electrically connects a first power source to a load;

a second switch comprising:

at least one second SSSD pair,

wherein the second switch electrically connects a second power source to the load; and

wherein the at least one first SSSD pair comprises a different type of switching devices relative to the at least one second SSSD pair;

wherein a controller is configured to selectively cycle the at least one first SSSD pair according to a first logic and the at least one second SSSD pair according to a second logic on and off to reduce inrush current to upstream devices when electrically coupling one of the first power source and the second power source to the load.

2. The transfer switch according to claim **1**, wherein the first switch comprises a plurality of first SSSD pairs,

wherein each of the first SSSD pairs are connected in anti-series.

3. The transfer switch according to claim **1**, wherein each switching device of the at least one first SSSD pair is selected from a group comprising SiC metal-oxide-semiconductor field-effect transistors (MOSFETs), insulated gate bipolar transistors (IGBTs), bipolar junction transistors (BJTs), junction-gate field effect transistors (JFETs), or any combinations thereof.

4. The transfer switch according to claim **3**, wherein each switching device of the at least one first SSSD pair comprises SiC MOSFETs.

5. The transfer switch according to claim **1**, wherein each switching device of the at least one first SSSD pair is selected from a group consisting of SiC MOSFETs, IGBTs, BJTs, JFETs, or any combinations thereof.

6. The transfer switch according to claim **1**, wherein the second switch comprises a plurality of second SSSD pairs,

wherein the plurality of second SSSD pairs are connected in anti-parallel.

7. The transfer switch according to claim **1**, wherein each switching device of the at least one second SSSD pair is selected from a group comprising SiC MOSFETs, IGBTs, BJTs, JFETs, silicon controlled rectifiers (SCRs), SCRs with paralleled resistive turn-off (RTO) circuits, or any combinations thereof.

8. The transfer switch according to claim **7**, wherein each switching device of the at least one second SSSD pair comprises:

SCRs with paralleled RTO circuits.

9. The transfer switch according to claim **1**, wherein each switching device of the at least one second SSSD pair is selected from a group consisting of SCRs, SCRs with paralleled RTO circuits, or any combinations thereof.

10. The transfer switch according to claim **1**, further comprising:

the controller, the controller comprises a processor and a non-transitory computer readable medium having stored thereon instructions executable by the processor to enable the controller to perform operations comprising:

send a first set of gate signals to the at least one first SSSD pair to open the first switch in response to detecting an abnormal condition at the first power source, and

after a time delay, sending a second set of gate signals to the at least one second SSSD pair to close the second switch and to connect the load to the second power source.

11. The transfer switch according to claim **10**, wherein the time delay for transferring the load between the first power source and the second power source is less than 10 msec.

12. The transfer switch according to claim **11**, wherein the controller transfers the load from the first power source to the second power source when the first power source and the second power source are out of phase by more than 30°.

13. A computer-implemented method for operating a transfer switch including a first switch and a second switch, the method comprising:

determining, by a controller, an abnormal condition at a first power source in electrical connection with the transfer switch;

sending, by the controller and based on a first logic, a first set of gate signals to the first switch to open at least one first SSSD pair in response to detecting the abnormal condition at the first power source; and

following a time delay, sending, by the controller and based on a second logic, a second set of gate signals to the second switch to close at least one second SSSD pair and to connect a load to a second power source,

wherein the at least one first SSSD pair comprises a different type of switching devices relative to the at least one second SSSD pair,

wherein the controller is configured to selectively cycle the at least one first SSSD pair according to the first logic and the at least one second SSSD pair according to the second logic on and off to reduce inrush current to upstream devices when electrically coupling one of the first power source and the second power source to the load.

14. The computer-implemented method according to claim **13**, wherein the delay time for transferring the load between the first power source and the second power source is less than 10 msec.

15. The computer-implemented method according to claim **13**, wherein the delay time for transferring the load is further configured to include when the first power source and the second power source are out of phase by more than 30°.

16. The computer-implemented method according to claim **13**, wherein the first logic is configured to operate switching devices of the at least one first SSSD pair, and the second logic is configured to operate switching devices of the at least one second SSSD pair.

17. The computer-implemented method according to claim **13**, wherein the first switch comprises a plurality of first SSSD pairs, and the second switch comprises a plurality of second SSSD pairs,

wherein each switching device in the plurality of first SSSD pairs are connected in anti-series and each switching device in the plurality of second SSSD pairs are connected in anti-parallel.

18. A system comprising:

a first power source;

a second power source;

an electrical load;

a transfer switch comprising:

a first switch comprising a plurality of first SSSD pairs connected in parallel between the first power source and the electrical load, and

a second switch comprising a plurality of second SSSD pairs connected in parallel between the second power source and the electrical load,

wherein the plurality of first SSSD pairs comprises different type of switching devices relative to the plurality of second SSSD pairs; and

a controller comprising:

a processor, and

a non-transitory computer readable medium having stored thereon instructions executable by the processor to enable the controller to perform operations comprising:

determine an abnormal condition at the first power source in electrical connection with the transfer switch,

send a first set of gate signals to the plurality of first SSSD pairs to open the first switch in response to detecting an abnormal condition at the first power source, and

after a time delay, sending a second set of gate signals to the plurality of second SSSD pairs to close the second switch and to connect the electrical load to the second power source,

wherein the controller is configured to selectively cycle the plurality of first SSSD pairs according to a first logic and the plurality of second SSSD pairs according to a second logic on and off to reduce inrush current to upstream devices when electrically coupling one of the first power source and the second power source to the electrical load.

19. The system according to claim **18**, wherein each switching device of the plurality of first SSSD pairs comprises SiC MOSFETs connected in anti-series, wherein each switching device of the plurality of second SSSD pairs comprises SCRs with paralleled RTO circuits connected in anti-parallel.

20. The system according to claim **18**, wherein the time delay for transferring the electrical load between the first power source and the second power source is less than 10 msec,

wherein the controller transfers the electrical load from the first power source to the second power source when the first power source and the second power source are out of phase by more than 30°.

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